

PAVANKALYAN NAGELLI

Software Development Engineer

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WORK EXPERIENCE

GLEAN TECHNOLOGIES, INC.

USA

Software Development Engineer (AI Platform & Automation)

May 2025 – Present

- Developed features for an **enterprise AI search platform** using Python and Retrieval-Augmented Generation (RAG), improving search relevance by **22% across internal knowledge systems**.
- Developed and maintained **LLM-based data ingestion pipelines** using Python and FastAPI to support vector-based semantic search.
- Improved performance of backend APIs by implementing asynchronous processing, reducing response time from **800ms to 250ms**.
- Automated data processing workflows using Python and n8n, improving efficiency and reducing manual intervention.
- Built internal tools using React and TypeScript to support monitoring of indexing workflows and prompt configurations.
- Collaborated with engineering and product teams to deliver **LLM-powered features** for enterprise applications.

LTIMINDTREE LIMITED

INDIA

Senior Software Engineer

Mar 2023–Jan 2024

- Improved API performance for Marsh & McLennan by implementing caching strategies and query optimization, reducing latency by **92%**.
- Contributed to development of **event-driven services** using Apache Kafka for real-time data processing across distributed systems.
- Developed backend services using Java (Spring Boot) and Python to support financial workflows and business logic.
- Participated in optimization and migration of microservices, improving system scalability and performance.
- Implemented Kafka-based messaging to enable reliable communication between distributed services.

Software Engineer

Mar 2021–Feb 2023

- Implemented user behavior tracking using Adobe Analytics, contributing to **18% improvement in user engagement**.
- Developed frontend features using React and Redux for financial applications.
- Supported backend development using Python and REST APIs for workflow automation.
- Contributed to digitization of workflows, reducing processing time from manual to near real-time execution.

EDUCATION

University of Central Missouri, MO, USA

December 2025

Master of Science in Computer Science

Guru Nanak Institute of Technology, HYD, India.

July 2021

Bachelor of Technology in Electronics & Communication Engineering

SKILLS & INTERESTS

Programming Languages: Python, Java, JavaScript, SQL

AI / Machine Learning: LLMs, RAG, LangChain, Hugging Face, Vector Embeddings, Semantic Search

Backend & APIs: FastAPI, REST APIs, Microservices, Spring Boot

Data & Messaging: PostgreSQL, MongoDB, Redis, Apache Kafka

Testing: PyTest, Unit Testing

Cloud & DevOps: AWS, Docker, CI/CD, Git

Frontend: React.js, TypeScript

PROJECTS

IoT Disaster Dam Management System (Arduino, Raspberry Pi, Python, GSM, Cloud Dashboard)

- Designed a smart dam-monitoring solution to send real-time flood alerts via sensors and GSM communication.

ACHIEVEMENTS

- Gained hands-on experience in developing and monitoring IoT-based sensor solutions at Adobe IoT Academy.
- Had the opportunity to be recognized as a Finalist in Smart India Hackathon.
- Served as an NSS Volunteer, contributing to community service and social initiatives.
- Supported campus outreach for Perplexity's Comet student tool.
- Contributed to digital content creation and event coverage for the UCM Student Newspaper. social awareness initiatives.